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One bad row? Keep going.

FORALL ... SAVE EXCEPTIONS finishes the batch, then tells you which rows failed.

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One error stops the whole FORALL

Without SAVE EXCEPTIONS, the first failing statement stops FORALL. The rows after it never run.

Unhandled, PL/SQL also rolls back the changes the earlier statements made.

Checked against Oracle documentation

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The setup

job is VARCHAR2(18). We add ' (Senior)', which is 9 characters, to every job.

deptno	job	length + 9
10	Clerk	14
20	Bookkeeper	19 > 18
30	Analyst	16

Row 20 will not fit.

Checked against Oracle documentation

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Plain FORALL

```
FORALL j IN depts.FIRST..depts.LAST
  UPDATE emp_temp
  SET job = job || ' (Senior)'
  WHERE deptno = depts(j);
```

```
ORA-12899: value too large
for column ... (actual: 19,
maximum: 18)
```

Statement 2 fails. Statement 3 (dept 30) **never runs**.

Checked against Oracle documentation

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Add SAVE EXCEPTIONS

```
FORALL j IN depts.FIRST..depts.LAST
  SAVE EXCEPTIONS
  UPDATE emp_temp
  SET job = job || ' (Senior)'
  WHERE deptno = depts(j);
```

A failing statement is saved, not raised. FORALL runs every statement, then raises one error: **ORA-24381**.

Checked against Oracle documentation

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Give ORA-24381 a name

```
dml_errors EXCEPTION;
PRAGMA EXCEPTION_INIT(
  dml_errors, -24381);
```

Now you can catch it with **WHEN dml_errors THEN** in the exception section.

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Read SQL%BULK_EXCEPTIONS

```
FOR i IN 1..SQL%BULK_EXCEPTIONS.COUNT
LOOP
  idx  := SQL%BULK_EXCEPTIONS(i).ERROR_INDEX;
  code := SQL%BULK_EXCEPTIONS(i).ERROR_CODE;
  msg  := SQLERRM(-code);
END LOOP;
```

ERROR_INDEX: which statement failed.
ERROR_CODE: the error number, positive, so negate it for SQLERRM.

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The result

The handler commits. Only the bad row is left unchanged.

deptno	job
10	Clerk (Senior)
20	Bookkeeper
30	Analyst (Senior)

ERROR_INDEX = 2, so depts(2) = 20 is the row to fix.

Checked against Oracle documentation

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Recap

- ✔ Plain FORALL stops at the first failing statement
- ✔ SAVE EXCEPTIONS runs them all, then raises ORA-24381
- ✔ SQL%BULK_EXCEPTIONS.COUNT = how many failed
- ✔ ERROR_INDEX = which one, ERROR_CODE = which error
- ✔ SQLERRM(-ERROR_CODE) gives the message

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Save this for your next bulk load.

Send it to a friend who writes PL/SQL, and follow @daily_sql_notes for a new SQL tip every day.